

Konstantinos Christopoulos

Robotics Software Engineer

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Computer Science & Engineering graduate (Integrated MSc) specializing in autonomous robot navigation, localization, and perception. Hands-on experience building real-time **ROS1** pipelines combining **LiDAR-based sensing**, sensor fusion (**Kalman filtering**), and motion planning, validated on a physical mobile robot platform. Also skilled in **Python/Java software engineering**

Education

Integrated Master (MSc) in Computer Science & Engineering 09/2018 – 02/2025 | University of Ioannina
Department of Computer Science & Engineering

- Thesis: Robot Navigation in Presence of People

Academic Projects & Research

Robot Navigation in Presence of People

- Designed a multi-step pedestrian avoidance framework for a Clearpath Jackal robot navigating dynamic, human-populated environments, tested against a Social-Force-Model pedestrian simulator (pedsim_ros).
- Built a real-time pipeline processing 3D point cloud data from a Velodyne VLP-16 LiDAR, using DBSCAN clustering for pedestrian detection.
- Applied Kalman filters for real-time state estimation, tracking and predicting pedestrian motion from LiDAR data.
- Developed a potential-field navigation algorithm for safe, collision-free robot movement around people and obstacles.
- Adapted a pre-built URDF/Xacro robot description (Clearpath Jackal) for Gazebo simulation.
- Validated the system through lab experiments and field trials on the physical robot, achieving robust real-time detection and avoidance with a modular ROS architecture.

Artificial Intelligence

- Developed a neural network in Java following OOP principles.
- Implemented and tested classification algorithms, evaluating model performance.

Full Stack Web Application Deployment

- Designed a full-stack application (Spring Boot backend, Thymeleaf frontend) with UML-based requirements analysis and JUnit/Mockito-tested features.

Skills

Robotics: ROS1/ROS Noetic (nodes, launch systems), Gazebo, RViz, URDF/Xacro, DBSCAN clustering, Kalman filtering & sensor fusion, potential-field motion planning
Programming Languages: Java, Python, C/C++, JavaScript, SQL, HTML/CSS
Machine Learning: NumPy, neural networks, classification algorithms
Testing: JUnit, Mockito
Backend & Frameworks: Spring Boot, Spring MVC, Spring Security, Hibernate/JPA, Thymeleaf, React
Tools & Workflow: Git, GitHub, Maven, Eclipse, VSCode, Ubuntu, LaTeX
Databases: MySQL

Professional Experience

Academic Tutor 03/2024 – 03/2025 | Ioannina, Greece
Self-employed

- Provided one-to-one tutoring to university students in programming and computer science courses.

Sales Assistant 07/2019 – Present | Ancient Olympia, Greece
Family-owned retail business

- Managed customer interactions and supported daily sales operations in a tourist retail environment during the summer season.

Languages

Greek

- Native

English

- ECCE B2 Level